Part A - Multiple Choice Questions (8*1=8 Marks)

1. Which of the following best defines an availability compromise?
   A. When data is modified without authorization
   B. When sensitive information is leaked to the public
   C. When systems or services become slow, unavailable, or unresponsive
   D. When digital signatures are invalid

2. Integrity in information security refers to:
   A. Ensuring only authorized access to sensitive data
   B. Ensuring accuracy, consistency, and trustworthiness of data
   C. Ensuring timely access to systems and services
   D. Encrypting all confidential information

3. The term "cracker" usually refers to someone who:
   A. Improves network security through testing
   B. Develop firewalls and antivirus software
   C. Breaks software encryption or licensing protections illegally
   D. Conducts authorized vulnerability assessments

4. Which of the following statements about "Grey Hat" hacker is correct?
   A. They hack only after obtaining permission and fix vulnerabilities
   B. They expose flaws without permission but not always for harmful reasons
   C. They hack systems with malicious intent to harm the system
   D. They are part of cybersecurity response teams

A

5. HTTPS ensures:

A. Open data exchange

B. Encrypted web communication

C. Anonymous access

D. File management

6. What is the main goal of penetration testing?

A. To permanently damage systems and gain money

B. To test specific security controls under real attack conditions

C. To monitor employee behavior and report it

D. To install malware on target systems

7. Which of the following describes passive reconnaissance?

A. Directly interacting with the target to collect information

B. Exploiting vulnerabilities in the target system

C. Collecting information without contacting the target

D. Installing backdoors to gain access

8. Wireless jamming leads to:

A. DoS by disrupting signals

B. Weak Encryption

C. Faster Wi-Fi speeds

D. Safe data transmission

3

Part B - Short Answers Questions (8 Marks)

1. During a weekly audit, an administrator discovers that several system log files were modified even though no maintenance was scheduled.

(2 Marks)

Answer the following questions:

a. Which integrity detection method would identify this incident in above scenario?

FIM File Integrity modification

b. What is the most likely cause of this issue?

Insiders

0.5

1.5

2. Analyze each scenario and identify security threats based on the type of attack:

(2 Marks)

Scenarios:	Type of attack:
A bank's IT security team notices a sudden spike in CPU usage across several employee workstations, even after work hours. Employees report that their computers are running unusually slow. Upon further investigation, **the team discovers unauthorized cryptocurrency mining software running in the background.** **Identify the type of Malware infected bank's computers?**	cryptojacking ~~(Malware)~~ 2
Attackers set up a rogue access point in Oman Avenues Mall **that mimics a legitimate wireless network**. Unknowingly some of the customers connect to the rogue access point, it is allowing attackers to intercept and manipulate their traffic. **Identify the type of wireless attack used by an attacker?**	Evil twin

3. The online banking system of Bank Muscat experiences cyberattacks. The bank wants a firewall that can track active connections to ensure that only valid responses are allowed.

(2 Marks)

Answer the following questions:

a. Which firewall type is most suitable for Bank Muscat?

NGFW

Next generation firewall X

b. What is the limitation of this type?

expensive and complex to deploy X

0

4. Fill in the blanks for the missing details that differentiate Ethical Hacking from Penetration Testing.

(2 Marks)

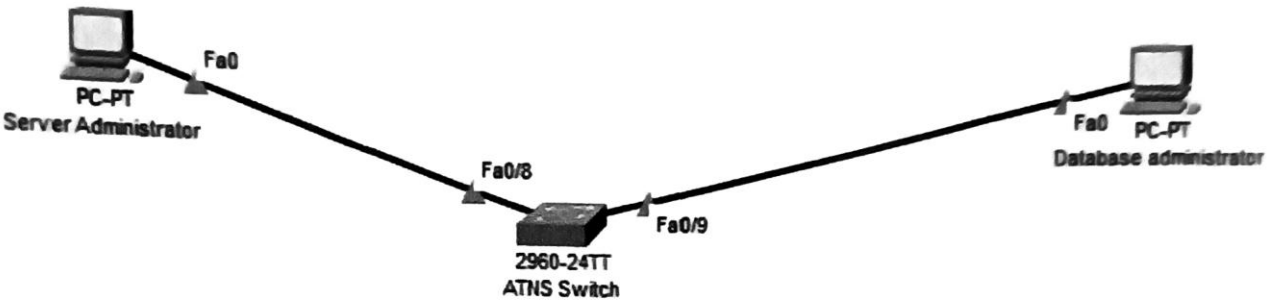
Aspect	Ethical Hacking	Penetration Testing
Scope	Entire system check	Targets specific systems, networks or applications
Purpose	Identify and fix vulnerabilities to improve overall security	testing a specific system to know how well the security is
Reporting	Comprehensive reports with detailed recommendations	Providing the weaknesses found on the system
Duration	Long-term	Short-term

2

Part C – Practical Assessment (4 Marks)

In the dynamic environment of Alpha-Tech Net Solutions (ATNS), the IT infrastructure is to be managed by two main administrators: Database Administrator and System Administrator. They monitor the workflow for their teams from their designated workstations connected to the company's network through a central switch.

Both of their workstations are connected to the network switch, which is configured with stringent port security measures to allow only authorized devices to connect. **You are asked to provide suitable commands in the network switch so that port security can be achieved.**



Details of the interfaces as below figure:

Device Name: ATNS Switch
Custom Device Model: 2960 IOS15
Hostname: Switch

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Down	1	--	0001.C996.AA01
FastEthernet0/2	Down	1	--	0001.C996.AA02
FastEthernet0/3	Down	1	--	0001.C996.AA03
FastEthernet0/4	Down	1	--	0001.C996.AA04
FastEthernet0/5	Down	1	--	0001.C996.AA05
FastEthernet0/6	Down	1	--	0001.C996.AA06
FastEthernet0/7	Down	1	--	0001.C996.AA07
FastEthernet0/8	Up	1	--	0001.C996.AA08
FastEthernet0/9	Up	1	--	0001.C996.AA09
FastEthernet0/10	Down	1	--	0001.C996.AA0A
FastEthernet0/11	Down	1	--	0001.C996.AA0B
FastEthernet0/12	Down	1	--	0001.C996.AA0C
FastEthernet0/13	Down	1	--	0001.C996.AA0D
FastEthernet0/14	Down	1	--	0001.C996.AA0E
FastEthernet0/15	Down	1	--	0001.C996.AA0F
FastEthernet0/16	Down	1	--	0001.C996.AA10
FastEthernet0/17	Down	1	--	0001.C996.AA11
FastEthernet0/18	Down	1	--	0001.C996.AA12
FastEthernet0/19	Down	1	--	0001.C996.AA13
FastEthernet0/20	Down	1	--	0001.C996.AA14
FastEthernet0/21	Down	1	--	0001.C996.AA15
FastEthernet0/22	Down	1	--	0001.C996.AA16
FastEthernet0/23	Down	1	--	0001.C996.AA17
FastEthernet0/24	Down	1	--	0001.C996.AA18
GigabitEthernet0/1	Down	1	--	0001.C996.AA19
GigabitEthernet0/2	Down	1	--	0001.C996.AA1A
Vlan1	Up	1	20.20.20.2/24	00D0.BC99.1DB6

Answer the following:

1. Enable port security on Fast Ethernet ports of the workstations in the above topology connected with ATNS Switch.

ATNS(config)# interface range Fa1/8 - 9

ATNS(config-if-range)# switchport port-security

(2 Marks)

2. Secure the ports so that the MAC address of a device is dynamically learned and added to the running configuration.

ATNS(config-if-range)# switchport port-security ~~mac-address~~ stickly

(1 Mark)

3. Verify that port security is enabled and the MAC addresses of PC-PT (server administrator) and PC-PT (Database administrator) were added to the running configuration.

ATNS(config-if-range)# crtl+z show run | port-security 0.5

show

begin interface

(1 Mark)

3.5